Dynamic Models: The Laplace Transforms

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1 What is a Laplace Transform

- Laplace Transform converts a signal in the time domain to a signal in the frequency domain, figure 1.

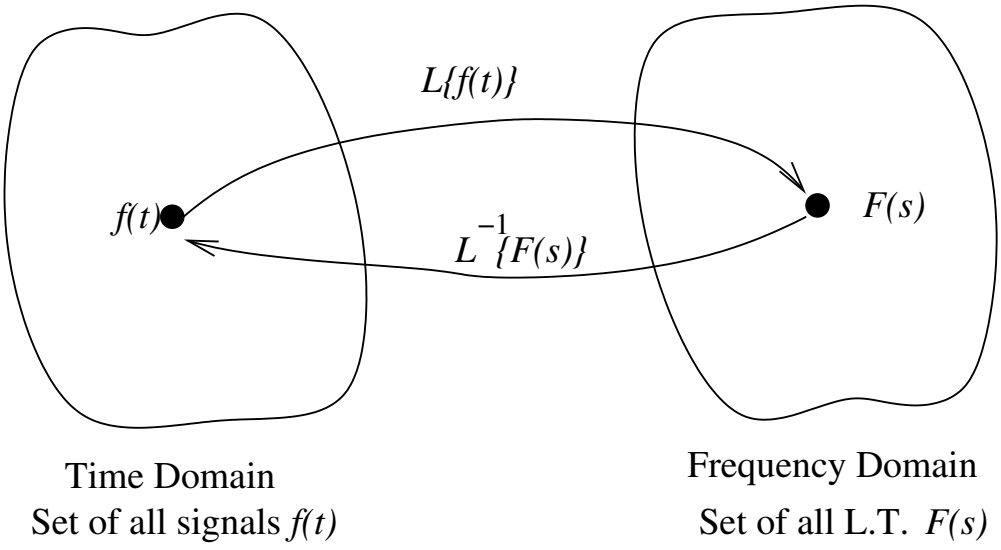


Figure 1: Laplace Transform is a one-to-one mapping between Time Domain and Frequency Domain.

1.1 One-sided Laplace transform integrals

- Formally, the Laplace transform is defined by the integrals:

$$F(s) = \mathcal{L}_{-}\{f(t)\} = \int_{0^{-}}^{\infty} f(t)\, e^{-st}\, dt \tag{1}$$

$$f(t) = \mathcal{L}_{-}^{-1}\{F(s)\} = \frac{1}{2\pi j}\int_{\sigma_c - j\infty}^{\sigma_c + j\infty} F(s)\, e^{st}\, ds\,, \quad \sigma_c > 0 \tag{2}$$

- The one-side Laplace transform applies most directly to controls analysis, because $f(t) = 0$ is assumed for $t < 0$.
- Equations (1) and (2) provide a formal definition for the Laplace transform. For Controls engineering it is the properties of the Laplace transform that are most important (section 2 of these notes).

2 Laplace transform properties and theorems (Franklin *et al.* section 3.1 (7th ed.))

Laplace transform properties and theorems are reviewed here without proof.

2.1 Superposition property

Because the integration in Eqn (1) is linear, and has the superposition property, the Laplace transform has the superposition property. Given:

$$f(t) \overset{\mathcal{L}}{\Longrightarrow} F(s)$$

and

$$g(t) \overset{\mathcal{L}}{\Longrightarrow} G(s)$$

then:

$$f(t)+g(t) \overset{\mathcal{L}}{\Longrightarrow} F(s)+G(s)$$

and

$$c_1 f(t)+c_2 g(t) \overset{\mathcal{L}}{\Longrightarrow} c_1 F(s)+c_2 G(s) \tag{3}$$

where $c_1, c_2 \in R$.

2.1.1 Scaling property

The scaling property is a special case of the superposition property. Given

$$f(t) \overset{\mathcal{L}}{\Longrightarrow} F(s)$$

then:

$$c f(t) \overset{\mathcal{L}}{\Longrightarrow} c F(s) \tag{4}$$

2.2 Differentiation property

The LT of the derivative of a signal is given by:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = \int_{0^-}^{\infty}\left(\frac{df(t)}{dt}\right)e^{-st}dt = -f\left(0^-\right) + sF(s) \tag{5}$$

Repeated applications Eqn (5) gives:

$$\mathcal{L}\left\{\frac{d^m f(t)}{dt^m}\right\} = s^m F(s) - s^{m-1} f\left(0^-\right) - s^{m-2}\frac{df\left(0^-\right)}{dt} - \cdots - \frac{d^{(m-1)} f\left(0^-\right)}{dt^{(m-1)}} \tag{6}$$

- In controls we ordinarily assume that the initial conditions are zero
  - This is a major point, when the Controls Engineer designs a controller, he or she usually does not know the precise signals the system will encounter.
  - That is, we wouldn't know any value other than zero for the initial conditions.

  Assuming the $f\left(0^-\right)$ and all its derivatives are zero in Eqn (6) gives the form ordinarily used:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) \tag{7}$$

  Or for higher derivatives:

$$\mathcal{L}\left\{\frac{d^m f(t)}{dt^m}\right\} = s^m F(s) \tag{8}$$

- Thus differentiating a signal is like multiplying its LT by s. This and the Integration Property are key reasons we use the Laplace Transform for controls analysis.

2.3 Integration property

Since the derivative of the integral of a function is the function, we find:

$$\mathcal{L}\left\{\int_0^t f(\tau)\,d\tau\right\} = \frac{1}{s}F(s) \tag{9}$$

2.4 Final value theorem (FVT)

Final Value Theorem: If $Y(s)$ represents the output of a stable system, then:

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) \tag{10}$$

Proof:

Recall

$$\mathcal{L}\left\{\frac{dy}{dt}\right\} = sY(s) - y\left(0^-\right) = \int_{0^-}^{\infty} e^{-st}\frac{dy}{dt}\,dt \tag{11}$$

In the limit as $s \to 0$, we have $e^{-st} \to 1$, so:

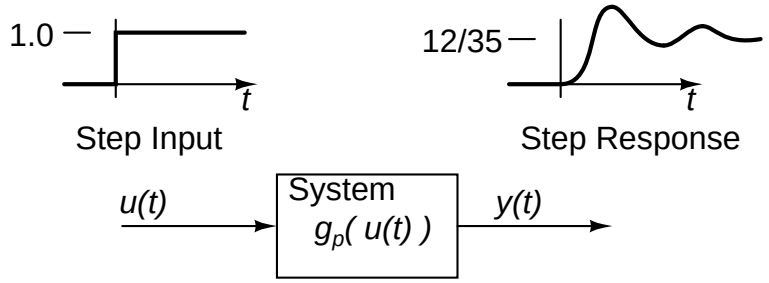
$$\lim_{s\to 0} sY(s) - y\left(0^-\right) = \lim_{s\to 0}\left[sY(s) - y\left(0^-\right)\right] = \lim_{s\to 0}\left(\int_{0^-}^{\infty} e^{-st}\frac{dy}{dt}\,dt\right) = \lim_{t\to\infty} y(t) - y\left(0^-\right)$$

When the $y\left(0^-\right)$ are canceled, this gives:

$$\lim_{s\to 0} sY(s) = \lim_{t\to\infty} y(t)$$

QED

2.4.1 Example application of the final value theorem to find steady state error



$$G_p(s) = \frac{Y(s)}{U(s)} = \frac{2s^2+10s+12}{s^2+12s+35}$$

Figure 2: System $G_p(s)$ with input $u(t) = 1^+(t)$.

- Note: for $u(t) = 1^+(t)$, $U(s) = \frac{1}{s}$.
- For the system of figure 2, $Y(s)$ is given by:

$$Y(s) = G_p(s)\,U(s) = \frac{2s^2+10s+12}{s^2+12s+35} \quad \frac{1}{s} \tag{12}$$

- *If the system is stable*, after enough time has gone by the output will have a steady value. To determine the steady state output apply the FVT:

$$y_{ss} = \lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \lim_{s\to 0} s\frac{2s^2+10s+12}{s^2+12s+35}\frac{1}{s}$$

- Canceling the s and $\frac{1}{s}$, and taking the limit gives y_{ss}, the steady-state value of $y(t)$, and the **steady-state error**:

$$y_{ss} = \lim_{s\to 0} \frac{2s^2+10s+12}{s^2+12s+35} = \frac{12}{35} \tag{13}$$

- Using $e(t) = r(t) - y(t)$,

$$e_{ss} = 1 - y_{ss} = \frac{23}{35} \tag{14}$$

2.4.2 DC gain of a system

Looking at the result of Eqn (13), we can see a general result that the DC gain of a system is given by the TF in the limit as $s \to 0$:

Define K_{DC}

$$K_{DC} = \lim_{s \to 0} G_p(s) \tag{15}$$

Applying this to the example:

$$K_{DC} = \lim_{s \to 0} \frac{2s^2 + 10s + 12}{s^2 + 12s + 35} = \frac{2 \cdot 0 + 10 \cdot 0 + 12}{0 + 12 \cdot 0 + 35} = \frac{12}{35}$$

Or as an even simpler result, writing

$$T(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0} \tag{16}$$

$$K_{DC} = \frac{b_0}{a_0} \tag{17}$$

where b_0 is the s^0 coefficient of the numerator, and a_0 is the s^0 coefficient of the denominator.

2.5 Questions on Laplace transform properties and theorems

Question:

1. List (without looking) the properties of the Laplace transform described here. Write a sentence or two describing each one.

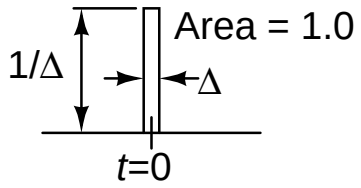
2. Which of these properties of the Laplace transform are required to develop a transfer function (TF) from an LT ?

3 Laplace transform pairs:

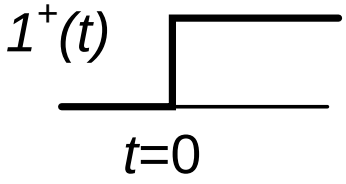
The essential Laplace Transform Pairs are given in table 1. The time domain functions are sometimes call elementary signals.

1. Impulse	$f(t) = \delta(t)$	$F(s) = 1.0$
2. Unit step function	$f(t) = 1^+(t)$	$F(s) = \frac{1}{s}$
3. Unit ramp function	$f(t) = t\ 1^+(t)$	$F(s) = \frac{1}{s^2}$
4. Decaying exponential	$f(t) = e^{-\alpha t}(t)$	$F(s) = \frac{1}{s+\alpha}$
5. Sinusoidal oscillation	$f(t) = B_c \cos(\omega t) + B_s \sin(\omega t)$	$F(s) = \frac{B_c s + B_s \omega}{s^2 + 0s + \omega^2}$
6. Oscillatory decay. exp.	$f(t) = e^{-\alpha t}(B_c \cos(\omega t) + B_s \sin(\omega t))$	$F(s) = \frac{B_c s + (B_c \alpha + B_s \omega)}{s^2 + 2\alpha s + \omega_n^2}$

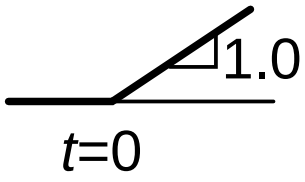
Table 1: Table of elementary Laplace transform pairs. $\left(\omega_n = \sqrt{\alpha^2 + \omega^2}\right)$



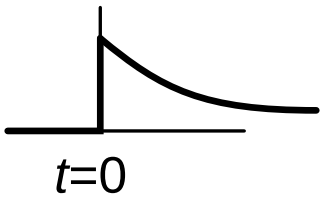
$\delta(t)$, impulse function



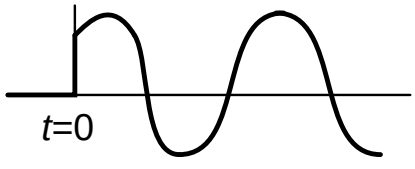
$1^+(t)$, unit step



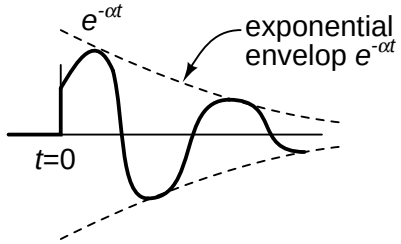
Unit ramp



Decaying exponential



A sinusoid



Oscillating decaying exponential

3.1 Notes on table 1

- Case 5 is just a special case of case 6.
- The forms for $F(s)$ for the Sinusoidal Oscillation and Oscillatory Decaying Exponential are a bit complex. It can't be helped.
- Notice that all of the Laplace transforms $F(s)$ are **rational polynomials**.

3.1.1 Rational polynomials and poles and zeros

- Consider the 6th case in table 1, with

$$y(t) = e^{-3t}\left(2\cos(\omega t) + 1.5\sin(\omega t)\right) \tag{18}$$

then

$$B_c = 2\,,\, B_s = 1.5\,,\, \alpha = -3\,,\, \omega = 4\,,\, \omega_n = \sqrt{\alpha^2+\omega^2} = 5$$

and

$$Y(s) = \frac{B(s)}{A(s)} = \frac{2s+12}{s^2+6s+25} \tag{19}$$

- Eqn (19) is an example of a rational polynomial: it is a ratio of a numerator and denominator polynomial in the Laplace variable, s.
- Rational polynomials have ***poles*** and ***zeros***

poles:	roots of the denominator polynomial
zeros:	roots of the numerator polynomial

- Considering

$$Y(s) = \frac{B(s)}{A(s)} = \frac{2s+12}{s^2+6s+25}$$

$B(s)$ and $A(s)$ as the numerator and denominator polynomials of $Y(s)$

Poles: $s_1 = -3 + j4$, and $s_2 = -3 - j4$

are the roots of $A(s) = s^2 + 6s + 25 = 0$.

Values s_1 and s_2 are the poles of $Y(s)$.

Note that the poles are complex, and are a ***complex conjugate*** pair.

Zeros: $s = -6$ is the root of $B(s) = 2s + 12 = 0$, it is the zero of $Y(s)$.

- The poles and zeros are plotted in a ***pole-zero map***, as in figure 3.

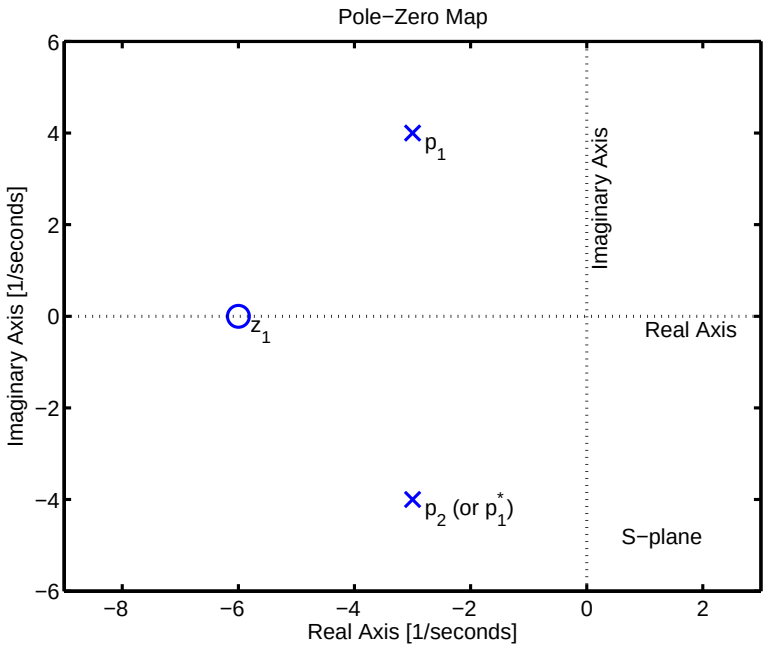


Figure 3: Pole-Zero map of $Y(s)$.

3.2 Impulsive function

The impulse is an interesting function, it is a short but intense burst with a total area under the curve of 1.0. A typical example is the impulsive jab on the gas pedal for which New York drivers are known. Other examples include:

- Force on a bullet when the cartridge fires;
- Force on an aircraft due to a sudden, brief gust.

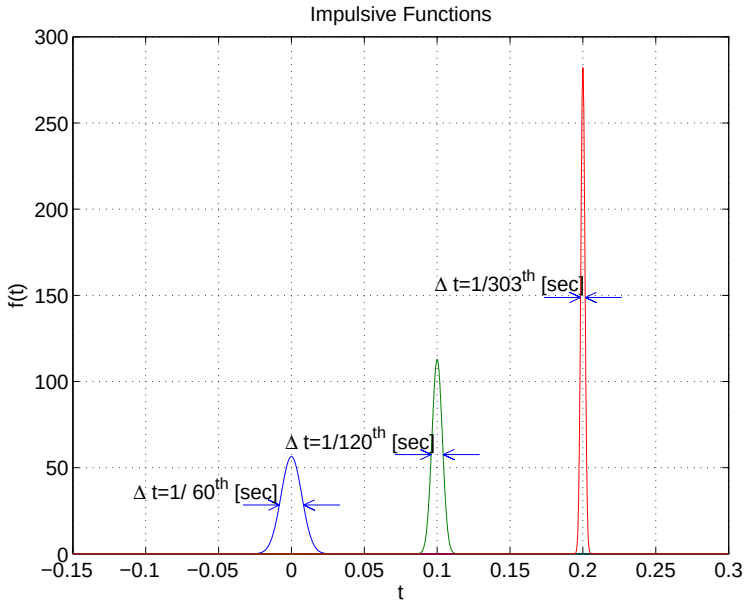


Figure 4: Three impulsive functions.

Three impulsive functions are illustrated in figure 4; the area under each is 1.0.

- The mathematical idealization of the impulse is given in the limit as:

$$\Delta t \to 0; \quad \text{and} \quad \max\left(\delta(t)\right) \to \infty \quad \text{with} \quad \int_{0^-}^{0^+} \delta(t)\, dt = 1.0 \qquad (20)$$

- For practical impulses, Δt should be short relative to the time constants of the dynamics of the system to which the impulse is applied; examples:
  - A few milliseconds for a bullet and rifle cartridge;
  - A second for a wind gust on a Boeing 747.

The Laplace transform of the impulse is given as:

$$F(s) = \int_{0^-}^{\infty} \delta(t)\, e^{-st}\, dt = \int_{0^-}^{0^+} \delta(t)\, e^{-st}\, dt = 1.0 \tag{21}$$

where the last equality follows from the facts that (in the limit as $\Delta t \to 0$):

- $\delta(t) = 0$ everywhere except the small interval from $t = 0^-$ to $t = 0^+$;
- $e^{-st} = 1$, on the interval from $t = 0^-$ to $t = 0^+$; and
- $\int_{0^-}^{0^+} \delta(t)\, dt = 1.0$ (definition of the impulsive function).

- A very good question to ask is:

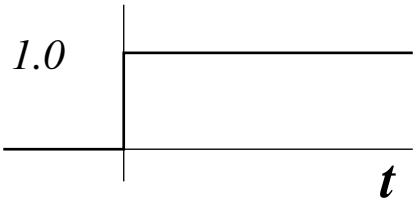
  "Why do we use the Laplace transform for control system analysis instead of another transform, such as the Fourier, Hartley, cosine or wavelet transforms ?"

- An important answer is that the Laplace transform is used because it leads us to the transfer function (whereas the Fourier transform, etc. do not).
  - The transform pair $f(t) = \delta(t) \longleftrightarrow F(s) = 1.0$ is part of the mathematical foundation for transfer functions.

3.3 Unit step function

- The unit step function is illustrated in figure 8. The notation used here for the unit step function is:[1]

$$f(t) = 1^{+}(t) \tag{22}$$



Illustrated

$$1^{+}(t) = \begin{cases} 0 & t < 0 \\ 1 & t \geq 0 \end{cases} \tag{23}$$

Equation

Figure 5: Unit step function.

- The unit step function serves two purposes:

1. To cut off functions for $t < 0$. For example the response of a system might be:

$$y(t) = 1^{+}(t)\, e^{-\alpha t}$$

where $y = 0$ for $t < 0$. (Generally, the Controls Engineer assumes no knowledge of what happened before $t = 0$).

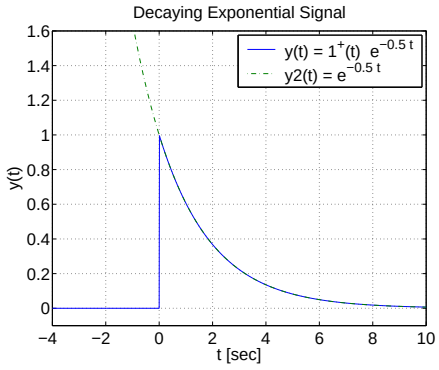


Figure 6: Exponential decay response, mathematically cut off at $t = 0$ by the unit step function.

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1. Franklin *et al.* use the notation $1(t)$, which can be confusing if the (t) argument is left off. Aström and others use the notation $1^{+}(t)$.

2. As a model input, to study the "step response" of the system

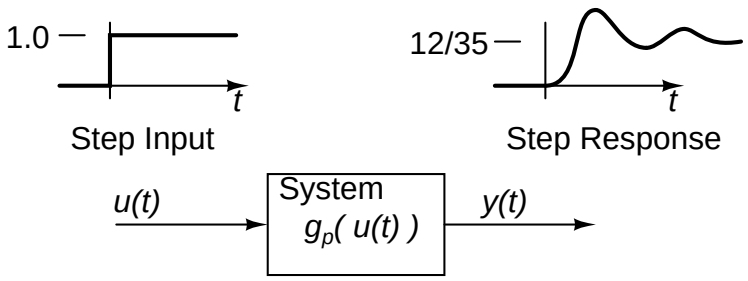


Figure 7: Study step response for overshoot, etc.

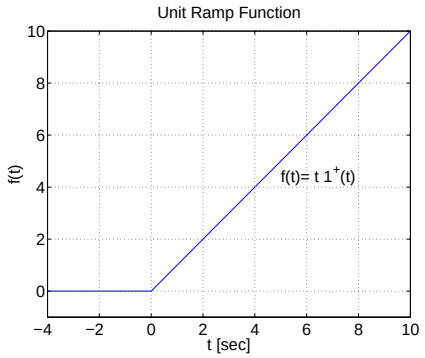
The Laplace transform of the unit step is:

$$\mathcal{L}\left\{1^{+}(t)\right\} = \frac{1}{s} \tag{24}$$

3.4 Unit ramp function

The unit ramp function is illustrated in figure 8. The unit ramp function is written:

$$f(t) = t\,1^{+}(t) \tag{25}$$



Illustrated

$$t\,1^{+}(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases} \tag{26}$$

Equation

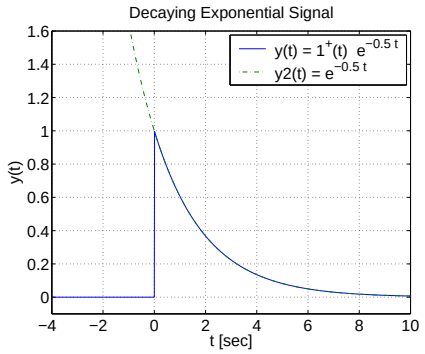
Figure 8: Unit step function.

The Laplace transform of the unit ramp signal is:

$$\mathcal{L}\left\{t\,1^{+}(t)\right\} = \frac{1}{s^2} \tag{27}$$

3.5 Decaying Exponential

A decaying exponential signal is illustrated in figure 6.



Illustrated

$$e^{-\alpha t}\,1^{+}(t) = \begin{cases} 0 & t < 0 \\ e^{-\alpha t} & t \geq 0 \end{cases} \qquad (28)$$

Equation

Figure 9: Decaying exponential signal.

The Laplace transform of the decaying exponential signal is:

$$\mathcal{L}\left\{e^{-\alpha t}\,1^{+}(t)\right\} = \frac{1}{s+\alpha} \qquad (29)$$

3.6 Sinusoidal oscillation

The sinusoidal oscillation is a special case of the exponentially decaying sinusoidal oscillation, which is considered in detail below.

- Considering Euler's formula:

$$e^{j\omega t} = \cos(\omega t) + j\sin(\omega t)\,, \qquad \sin(\omega t) = \frac{1}{2j}\left(e^{j\omega t} - e^{-j\omega t}\right)$$

- Gives:

$$\mathcal{L}\{\sin(\omega t)\} = \frac{1}{2j}\int_{0^-}^{\infty}\left(e^{(j\omega - s)t} - e^{-(j\omega + s)t}\right)dt = \frac{\omega}{s^2+\omega^2} \qquad (30)$$

- Likewise:

$$\mathcal{L}\{\cos(\omega t)\} = \frac{s}{s^2+\omega^2} \qquad (31)$$

3.7 Decaying sinusoidal oscillation

- The decaying sinusoidal oscillation is given by:

$$y(t) = e^{-\alpha t} A \cos(\omega t + \phi) \tag{32}$$

and is seen in figure 10.

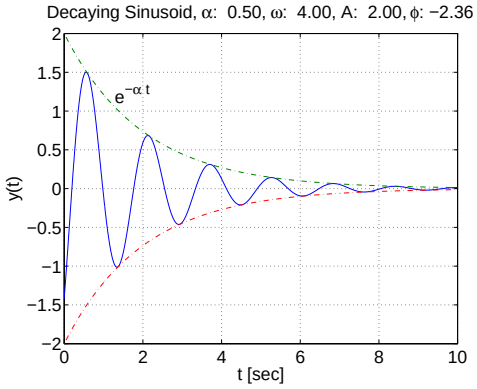


Figure 10: Decaying Sinusoid.

- The decaying sinusoid is a common system response. It is the most complicated signal and Laplace transform we will work with this semester.

  We will derive multiple forms for $y(t)$ and $Y(s)$ in a later lecture. For now let's consider one form.

- Recall that

$$\mathcal{L}\left\{e^{-\alpha t}\right\} = \frac{1}{s+\alpha}, \quad \text{so} \qquad \mathcal{L}^{-1}\left\{\frac{C_1}{s-p_1}\right\} = y(t) = C_1 e^{p_1 t} \tag{33}$$

- What if p_1 and C_1 are complex ? Introducing complex terms

$$\begin{aligned} C_1 &= c_r + j c_j \\ p_1 &= -\alpha + j\omega \end{aligned}$$

- With the complex terms,

$$V(s) = \frac{C_1}{s-p_1} \quad \Longrightarrow \quad v(t) = \left(c_r + j c_j\right) e^{(-\alpha + j\omega)t} \tag{34}$$

Decaying sinusoidal oscillation (continued)

- Now put complex $V(s)$ together with its complex conjugate,

$$\begin{aligned} Y(s) &= V(s)+V^*(s)=\frac{C_1}{s-p_1}+\frac{C_1^*}{s-p_1^*} \\ y(t)=\mathcal{L}^{-1}\{Y(s)\} &= \mathcal{L}^{-1}\left\{\frac{C_1}{s-p_1}+\frac{C_1^*}{s-p_1^*}\right\} \\ &\Longrightarrow \\ y(t) &= \left(c_r+jc_j\right)e^{(-\alpha+j\omega)t}+\left(c_r-jc_j\right)e^{(-\alpha-j\omega)t} \end{aligned} \tag{35}$$

- The exponential has real and imaginary parts, which we can dissect by:

$$e^{(-\alpha+j\omega)t} = e^{-\alpha t}\,e^{j\omega t} \tag{36}$$

- Putting Eqn (35) together with (36) gives:

$$y(t)=e^{-\alpha t}\left(\left(c_r+jc_j\right)e^{j\omega t}+\left(c_r-jc_j\right)e^{-j\omega t}\right) \tag{37}$$

- Using Euler's Formula

$$e^{j\omega t}=\cos(\omega t)+j\sin(\omega t) \tag{38}$$

then

$$\begin{aligned} y(t) &= e^{-\alpha t}\ \left(c_r\cos(\omega t)-c_j\sin(\omega t)+j\left(c_r\sin(\omega t)+c_j\cos(\omega t)\right)\right. \\ &\quad \left.+c_r\cos(\omega t)-c_j\sin(\omega t)-j\left(c_r\sin(\omega t)+c_j\cos(\omega t)\right)\right) \end{aligned}$$

$$y(t) = e^{-\alpha t}\ \left(2c_r\cos(\omega t)-2c_j\sin(\omega t)\right) \tag{39}$$

- Eqn (37) has complex conjugate terms, so all the imaginary parts must drop out !

- Eqn (32) gives a polar form for $y(t)$, while (39) gives a rectangular form.

$$y(t) = e^{-\alpha t}A\cos(\omega t+\phi) \qquad \text{(Eqn (32), repeated)}$$

Decaying sinusoidal oscillation (continued)

- The forms for $y(t)$ are tied together (and to $Y(s)$) by:

$$y(t) = e^{-\alpha t} A\cos(\omega t + \phi) = e^{-\alpha t}\left(B_c\cos(\omega t) + B_s\sin(\omega t)\right)$$

with

$$\begin{aligned} B_c &= 2c_r \\ B_s &= -2c_j \end{aligned} \tag{40}$$

and

$$\begin{aligned} A &= \sqrt{B_c^2 + B_s^2} \\ \phi &= \text{atan2}(-B_s, B_c) \end{aligned} \tag{41}$$

- With residue $C_1 = c_r + jc_j$, $Y(s)$ is given by:

$$Y(s) = \frac{C_1}{s-p_1} + \frac{C_1^*}{s-p_1^*} = \frac{c_r + jc_j}{s+\alpha-j\omega} + \frac{c_r - jc_j}{s+\alpha+j\omega} \tag{42}$$

$$= \frac{b_1 s + b_2}{s^2 + a_1 s + a_2} \tag{43}$$

  where the complex form of Eqn (42) can be tied to the all-real form of Eqn (43) by multiplying out the complex conjugate terms to put them over a common denominator.

- Thus, the complex signal of figure 10, which is very common in control systems, corresponds to a complex pole pair at:

$$p_1 = -\alpha + j\omega, \qquad p_2 = -\alpha - j\omega$$

  and each pole has a complex residue:

$$C_1 = c_r + jc_j, \qquad C_2 = c_r \pm -jc_j$$

4 Inverse Laplace transform (Franklin *et al.* sec. 3.1.5 (7th ed.))

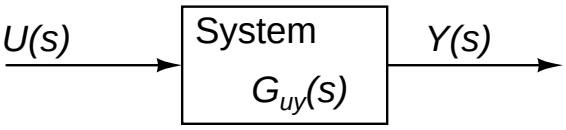
Broadly, the Controls Engineer uses the Laplace transform in two ways.

1. As the foundation for the transfer function
2. To analyze the response of the system to standard test inputs

- The connection between Laplace transforms and the transfer function we have seen
- Analyzing the response to standard test inputs will require the inverse Laplace transform

4.1 Inverse Laplace transform by Partial Fraction Expansion

- Notice that the Laplace transforms of all of the elementary signals in table 1 are rational polynomials. When we form an output LT:



$$Y(s) = G_{uy}(s)\, U(s)$$

  - If the input LT, $U(s)$, is a rational polynomial,
  - And the TF, $G_{uy}(s)$, is a rational polynomial

  $\Longrightarrow$ Then $Y(s)$ will be a rational polynomial !!!

- Any rational polynomial (that is, any $Y(s)$) can be:
  - Broken down into components which appear in the table of elementary signals.
  - This 'breaking-down' of $Y(s)$ into components is called ***Partial Fraction Expansion***. PFE

4.1.1 Partial Fraction Expansion example

Given the system and input:

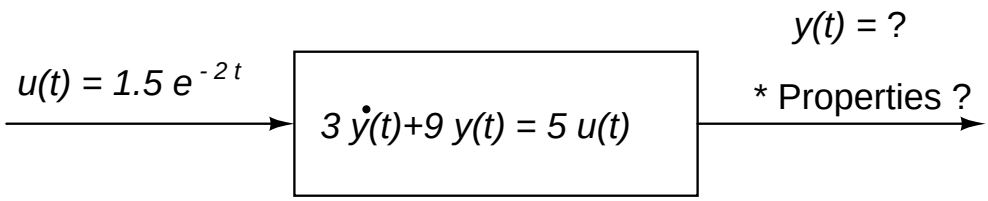


Figure 11: System with input signal $u(t)$.

- Find $Y(s)$ and $y(t)$

Solving for $Y(s)$:

$$Y(s) = G_{uy}(s)\,U(s)$$

with:

$$U(s) = 1.5\frac{1}{s+2} \quad , \quad G_{uy}(s) = \frac{5}{3s+9} \quad \text{then} \quad Y(s) = \frac{7.5}{3s^2+15s+18} \quad (44)$$

Solving for $y(t)$:

- Ultimately, we solve for $y(t)$ by looking up the time-domain functions in table 1.
  - Table 1 does not have a quadratic entry, of the form of $Y(s)$ in Eqn (44).
  - However, there is a first-order term of the form

$$C\,e^{-\alpha t} \quad \stackrel{\mathcal{L}}{\Longrightarrow} \quad \frac{C_i}{s+\alpha_i} \tag{45}$$

  - If we could factor (break-down) $Y(s)$ into the form

$$Y(s) = \frac{C_1}{s+\alpha_1} + \frac{C_2}{s+\alpha_2} \tag{46}$$

    then $y(t)$ is given as

$$y(t) = C_1\,e^{\alpha_1 t} + C_2\,e^{\alpha_2 t} \tag{47}$$

- We put $Y(s)$ in factored form by factoring the denominator polynomial:

$$Y(s) = \frac{B(s)}{A(s)} = \frac{7.5}{3s^2+15s+18} = \frac{2.5}{s^2+5s+6} = \frac{2.5}{(s+2)(s+3)} \tag{48}$$

  - Where in the first step we write $Y(s)$ in ***monic*** form, that is, the leading coefficient in the denominator is 1.0 .
  - In the second step we use the poles (roots of $A(s)$) to factor the denominator, that is we use

$$s^2+5s+6 = (s+2)(s+3) \tag{49}$$

- $Y(s)$ as given,

$$Y(s) = \frac{2.5}{(s+2)(s+3)} = 2.5\,\frac{1}{(s+2)}\,\frac{1}{(s+3)} \tag{50}$$

  is in ***product of factors*** form.

- The same rational polynomial (now considered just a mathematical function) can be written in ***sum of factors*** form

$$Y(s) = \frac{C_1}{s+2} + \frac{C_2}{s+3} \tag{51}$$

- Eqn (51) is the ***partial fraction expansion*** of $Y(s)$. In general, the ***PFE*** is $Y(s)$ expanded as a sum of terms that appear in table 1.

- The unknown coefficients C_1 and C_2 are called the ***residues***.
  (The term comes from the field of mathematics concerned with properties complex functions, such as $Y(s)$)

- Finding the residues will enable us to develop theoretical results and design tools for control system analysis.

  – In section 5 we will rigorously develop different methods for partial fraction expansion (each will lead us to certain theoretical results)

  – Here, let's carry on with the example, and see one way to find $y(t)$

- Finding C_1 and C_2 by the ***"plug and chug"*** method
  - To find $y(t)$, $Y(s)$ should be expressed in terms that appear in the table of Laplace transforms (also on the front flap of Franklin *et al.*).
  - $Y(s)$ in Eqn (44) does not appear in table 1, however we can factor $Y(s)$ into a sum of first order terms that do appear in the table. Starting with

$$Y(s) = \frac{2.5}{(s+2)(s+3)} = \frac{C_1}{(s+2)} \frac{C_2}{(s+3)} \tag{52}$$

  - We can find the residues by multiplying the sum terms back together:

$$\begin{aligned} Y(s) &= \frac{C_1}{s+2} + \frac{C_2}{s+3} = \frac{C_1(s+3) + C_2(s+2)}{(s+2)(s+3)} \\ &= \frac{(C_1+C_2)s + 3C_1 + 2C_2}{(s+2)(s+3)} \end{aligned} \tag{53}$$

  - Equating Eqns (52) and (53)

$$Y(s) = \frac{0s+2.5}{(s+2)(s+3)} = \frac{(C_1+C_2)s + 3C_1 + 2C_2}{(s+2)(s+3)}$$

    * We see that the denominators match.
    * Matching the numerators gives:

$$0s + 2.5 = (C_1+C_2)s + 3C_1 + 2C_2 \tag{54}$$

    * Considering that the numerators must match for all values of s, the coefficients must match for each power of s, giving two equations in two unknowns:

$$\begin{aligned} s^1 \text{ term:} \quad 0 &= C_1 + C_2 \\ s^0 \text{ term:} \quad 2.5 &= 3C_1 + 2C_2 \end{aligned} \tag{55}$$

– Solving for C_1 and C_2 gives

$$\begin{aligned} C_1 &= +2.5 \\ C_2 &= -2.5 \end{aligned} \tag{56}$$

- Plugging in C_1 and C_2

$$Y(s) = \frac{+2.5}{s+2} + \frac{-2.5}{s+3} \tag{57}$$

- From the table of Laplace transforms we find:

$$f(t) = C e^{-\alpha t} \longleftrightarrow F(s) = \frac{C}{s+\alpha} \tag{58}$$

And so Eqn (57) has the inverse Laplace transform

$$y(t) = 2.5\,e^{-2t} - 2.5\,e^{-3t}\,, \quad t \geq 0 \tag{59}$$

which is plotted in figure 12.

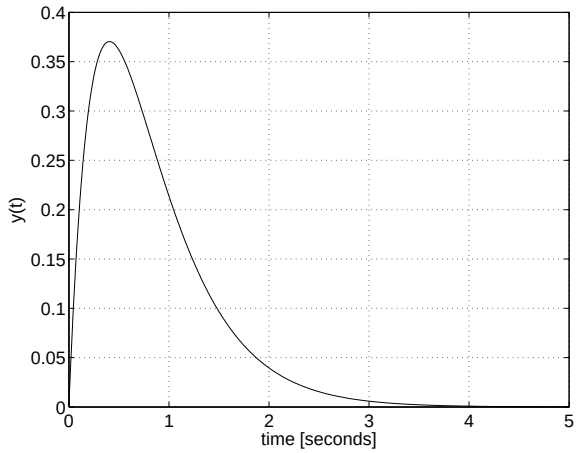


Figure 12: Response of system of figure 11.

- The ability to theoretically determine the response, as in figure 12, gives us the ability to study the stability, speed, magnitude of response and other control system characteristics of the system of figure 11.

4.2 Partial fraction expansion, general form

- Starting with a general LT:

$$Y(s) = \frac{b_m s^m + \cdots + b_1 s + b_0}{a_n s^n + \cdots + a_1 s + a_0} \tag{60}$$

This is rewritten in sum-of-terms form, with first and second order terms:

$$Y(s) = \frac{C_1}{s - p_1} + \cdots + \frac{C_c s + C_s}{s^2 + 2\alpha s + \omega_n^2} \tag{61}$$

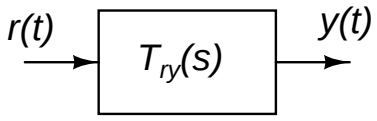
  - The p_i are the real roots of the characteristic equation
  - The α and ω_n come from the the complex roots of the characteristic equation
  - The C_i are the ***residues***.

- Solving for the partial fraction expansion requires finding the residues
  - By the plug and chug method
  - By the cover-up method
  - By a graphical method we'll see in the next section
  - By the Matlab `residue()` command.

4.3 Using inverse Laplace transform to explore system response to standard input signals

An important challenge of controls is to connect performance specifications to controller design choices.

- Most often, controller specifications are made in terms of the system response to standard test signals. Example standard test signals are:
- Step input
  - Rise time, Settling time, Percent overshoot, Steady state error
- Ramp input
  - Steady state error
- Sinusoidal input
  - Bandwidth, Phase margin, Gain margin
- The inverse Laplace transform mathematically connects the transfer function to the system output:



$$r(t) \quad \text{Standard test signal} \tag{62}$$

$$Y(s) = T_{ry}(s)\, R(s) \quad , \quad Y(s) \text{ given using the TF} \tag{63}$$

$$y(t) = \mathcal{L}^{-1} Y(s) \quad , \quad \mathcal{L}^{-1} \text{ connects } T_{ry}(s) \text{ to the output} \tag{64}$$

5 Three methods to determine the residues of partial fraction expansion

Three methods to determine the residues, C_i, are:

1. Plug and chug method,
2. Cover-up method, and
3. Graphical method.

5.1 Method 1: Plug in and solve (Plug and Chug):

The Plug and Chug method is shown by example. Suppose

$$Y(s) = \frac{3}{s^2+5s+6} = \frac{3}{(s+2)(s+3)} \tag{65}$$

Writing $Y(s)$ in sum-of-factors form, we need to solve for residues C_1 and C_2:

$$Y(s) = \frac{0s+3}{(s+2)(s+3)} = \frac{C_1}{(s+2)} + \frac{C_2}{(s+3)}$$

Putting the right-hand side over the common denominator gives:

$$\frac{C_1(s+3)+C_2(s+2)}{(s+2)(s+3)} = \frac{(C_1+C_2)s+(3C_1+2C_2)}{\Delta(s)} = Y(s) \tag{66}$$

Plug and chug (continued)

From the last equality in Eqn (66):

$$\frac{(C_1+C_2)\,s+(3C_1+2C_2)}{\Delta(s)} = \frac{0\,s+3}{\Delta(s)}$$

The essential steps of the Plug and Chug method is to equate the coefficients of the powers of s, giving:

$$\begin{aligned} C_1+C_2 &= 0 \\ 3\,C_1+2\,C_2 &= 3 \end{aligned}$$

Which gives:

$$C_1 = 3\,; \qquad C_2 = -3$$

And we find:

$$Y(s) = \frac{3}{s+2} + \frac{-3}{s+3}\,, \qquad y(t) = 3\,e^{-2t} - 3\,e^{-3t} \tag{67}$$

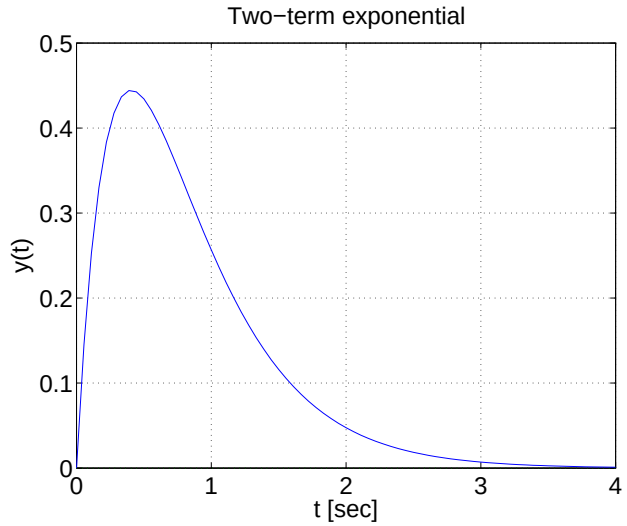


Figure 13: Plot of $y(t)$.

Notes:

- The amount of algebra grows rapidly with the number of terms.
- Can you see why Eqn (67) gives the curve of figure 13 ?

5.2 Method 2: The "Cover-up" Method

- Starting with the LT in product-of-factors form:

$$Y(s) = K_{rlg} \frac{\Pi_{i=1}^{m} (s - z_i)}{\Pi_{i=1}^{n} (s - p_i)}$$

- For example:

$$Y(s) = \frac{3s^2 + 18s + 24}{s^3 + 4s^2 + 3s + 0} = 3 \frac{s^2 + 6s + 8}{s^3 + 4s^2 + 3s + 0} \tag{68}$$

- Find poles and zeros, in Matlab

```
>>Zeros = roots( [1, 6, 8 ] )
  Zeros = [ -2, -4]
>> Poles = roots( [1, 4, 3, 0] )
  Poles = [ 0, -1, -3]
```

Put in product-of-factors form:

$$Y(s) = 3 \frac{(s+2)(s+4)}{(s+0)(s+1)(s+3)} \tag{69}$$

with:

$$K_{rlg} = 3; \quad p_1 = 0; \quad p_2 = -1; \quad p_3 = -3; \quad z_1 = -2; \quad z_2 = -4$$

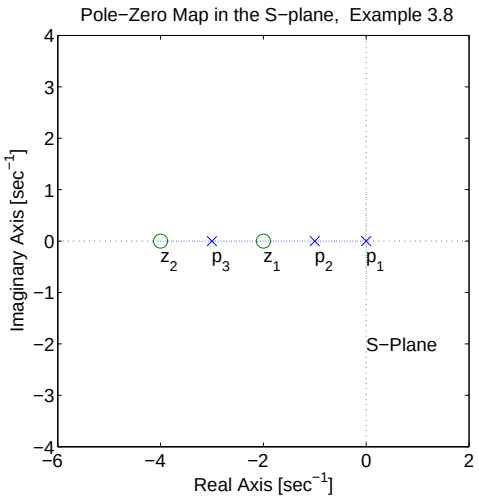


Figure 14: Pole-zero map of rational polynomial $Y(s)$.

- Express the 3rd order LT in sum-of-terms form:

$$Y(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \frac{C_3}{s-p_3} \tag{70}$$

  - p_i : Poles
  - C_i : Residues

- ***"Cover-up"*** a pole one at a time with a zero, by multiplying by a zero at that pole location. To "cover-up" pole p_1, multiply $Y(s)$ by $(s-p_1)$:

$$\begin{aligned} Y(s)\,(s-p_1) = \;& C_1\tfrac{s-p_1}{s-p_1} + C_2\tfrac{s-p_1}{s-p_2} + C_3\tfrac{s-p_1}{s-p_3} \\ &= C_1 + C_2\tfrac{s-p_1}{s-p_2} + C_3\tfrac{s-p_1}{s-p_3} \end{aligned} \tag{71}$$

- Evaluate this at $s = p_1$

$$Y(s)\,(s-p_1)\Big|_{s=p_1} = C_1 + C_2\frac{0}{s-p_2} + C_3\frac{0}{s-p_3} = C_1 \tag{72}$$

- Considering the pole at $p_1 = 0$

$$C_1 = Y(s)\,(s+0)\Big|_{s=0} = 3\,\frac{(s+2)\,(s+4)}{(s+0)\,(s+1)\,(s+3)}\,(s+0)\,\Big|_{s=0} = 3\,\frac{2\cdot 4}{1\cdot 3} = 8$$

  Giving the first residue:

$$Y(s) = \frac{8}{s} + \frac{C_2}{s+1} + \frac{C_3}{s+3}$$

- For C_2 :

$$C_2 = Y(s)(s+1)\Big|_{s=-1} = 3\frac{(s+2)(s+4)}{s(s+1)(s+3)}(s+1)\Big|_{s=-1} = 3\frac{1\cdot 3}{-1\cdot 2} = -\frac{9}{2}$$

$$\Longrightarrow \quad Y(s) = \frac{8}{s} + \frac{-9/2}{s+1} + \frac{C_3}{s+3}$$

- For C_3 :

$$C_3 = Y(s)(s+3)\Big|_{s=-3} = 3\frac{(s+2)(s+4)}{s(s+1)(s+3)}(s+3)\Big|_{s=-3} = 3\frac{-1\cdot 1}{-3\cdot -2} = -\frac{1}{2}$$

Thus
$$Y(s) = \frac{8}{s} - \frac{9/2}{s+1} - \frac{1/2}{s+3} \tag{73}$$

$$y(t) = 8\,1^{+}(t) - \frac{9}{2}e^{-t} - \frac{1}{2}e^{-3t} \tag{74}$$

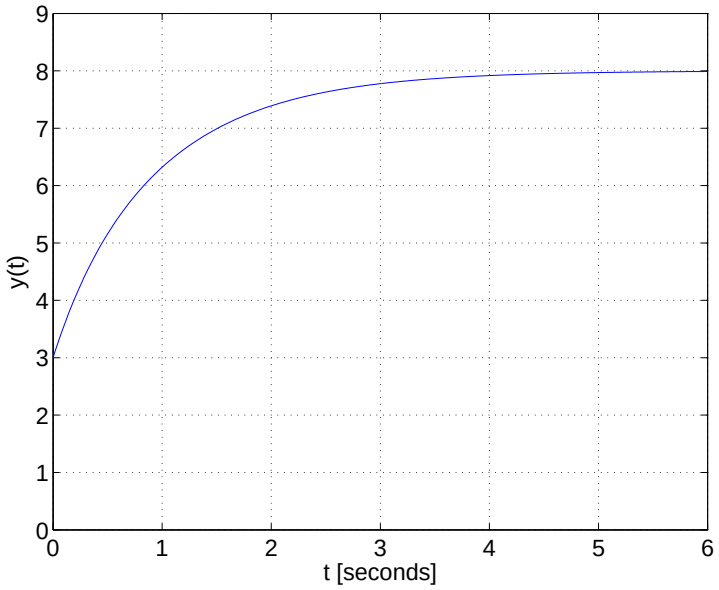


Figure 15: Signal $y(t)$ given by $Y(s)$ using partial fraction expansion and the cover-up method.

- We call this the *"cover-up"* method because we cover-up each pole in succession with a zero.

- The Matlab step and impulse commands use partial fraction expansion (and quite possibly the cover-up method) to compute the responses of linear systems.

5.3 Method 3: Graphical residue method

In general, the numerator and denominator of $Y(s)$ can be factored to put $Y(s)$ into product-of-factors form:

$$Y(s) = K_{rlg} \frac{(s-z_1)(s-z_2)\cdots}{(s-p_1)(s-p_2)\cdots} \tag{75}$$

- This is a function of s, plug in any value of s and Eqn (75) evaluates to a number. For example, with the point $s = 2$:

$$Y(s)\Big|_{s=2} = K_{rlg} \frac{(2-z_1)(2-z_2)}{(2-p_1)(2-p_2)(2-p_3)} = 3\frac{4\cdot 6}{2\cdot 3\cdot 5} = \frac{12}{5} \tag{76}$$

- Double check by directly plugging in :

$$Y(s)\Big|_{s=2} = \frac{3(2)^2 + 18(2) + 24}{(2)^3 + 4(2)^2 + 3(2)} = \frac{72}{30} = \frac{12}{5}$$

5.3.1 Calculating $Y(s)$ at $s = p_i$, and thus the residues

- Considering Eqn (75), the LT, evaluated for any value $s = s_0$, is given by:

$$Y(s)\Big|_{s=s_0} = K_{rlg} \frac{\text{Product of Vectors from zeros to } s_0}{\text{Product of Vectors from poles to } s_0} \tag{77}$$

The vectors in Eqn (77) are vectors to value s_0 , as seen in figure 16.

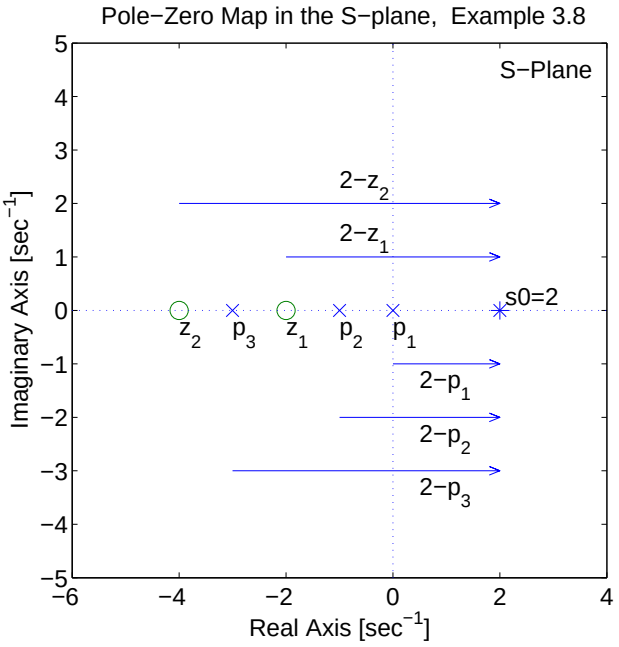


Figure 16: Vectors of the graphical residue calculation.

- In the cover-up method, we cover-up pole p_i with a zero and compute

$$C_i = Y(s)(s - p_i)\Big|_{s=p_i}$$

- The graphical residue method follows the same steps, graphically evaluating $Y(s)(s - p_i)\Big|_{s=p_i}$ to compute the residue, by selecting $s_0 = p_i$

$$C_i = Y(s)(s - p_i)\Big|_{s=p_i} = K_{rlg} \frac{\text{Product of Vectors from zeros to } i^{th} \text{ pole}}{\text{Product of Vectors from other poles to } i^{th} \text{ pole}}$$

Graphical residue method (continued)

- For example, for $Y(s)$ above,

$$Y(s) = 3\,\frac{s^2+6s+8}{s^3+4s^2+3s+0} \tag{78}$$

$K_{rlg} = 3.0$. Computing the residue at p_2, is illustrated in figure 17.

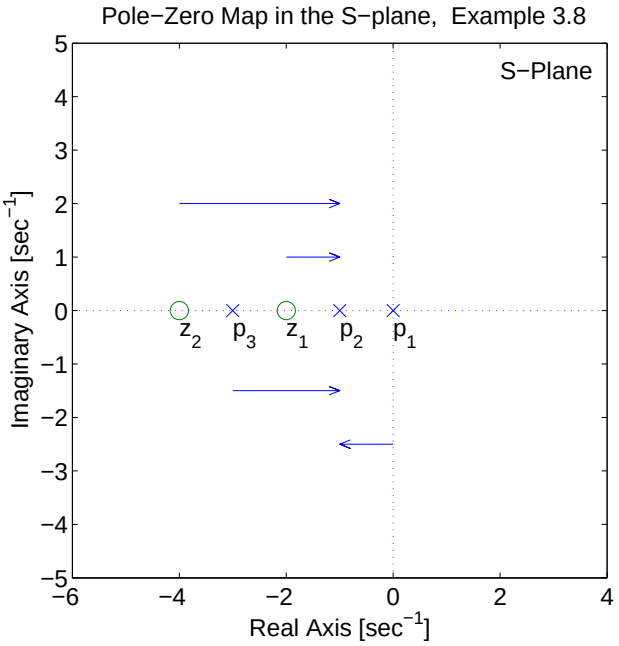


Figure 17: Graphical computation of the residue at pole p_2.

- Calculating, gives

$$C_2 = K_{rlg}\,\frac{\Pi\,\text{Vecs from zeros to } p_2}{\Pi\,\text{Vecs from other poles to } p_2} = 3\,\frac{1\cdot 3}{(-1)\cdot 2} = \frac{-9}{2}$$

just as we got with the cover-up method.

- The graphical method is the basis of root-locus design, a powerful controls design tool and the subject of chapter 5.

5.3.2 Graphical residue method and complex poles and zeros

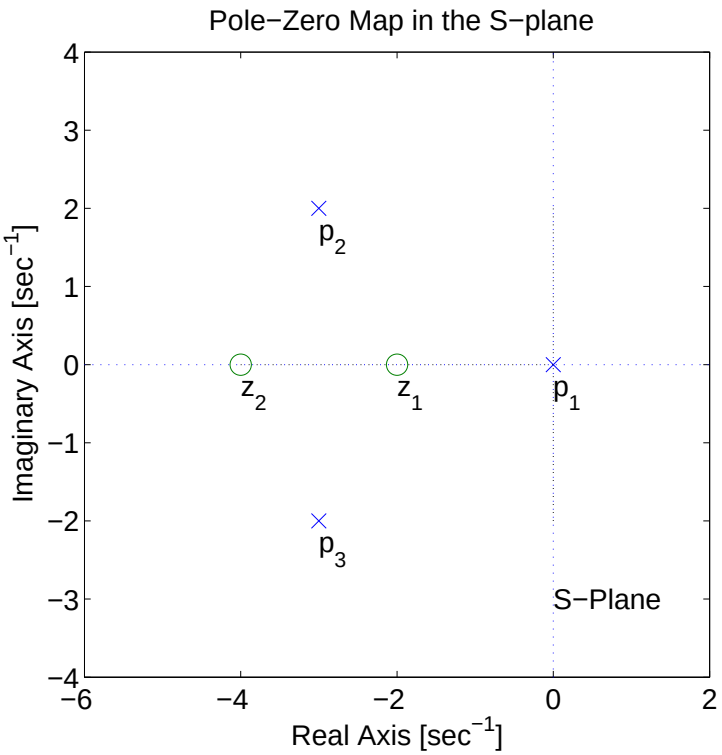
- Consider a step input applied to a system with transfer function

$$G_{uy}(s) = \frac{2\,s^2 + 12\,s + 16}{s^2 + 6\,s + 13}$$

- The Laplace transform of the output is given as:

$$Y(s) = G_{uy}(s)\,U(s) = \frac{2s^2 + 12s + 16}{s^2 + 6s + 13}\,\frac{1}{s} = 2\,\frac{(s+2)(s+4)}{s\,(s^2 + 6s + 13)} \tag{79}$$

where the last expression is in product-of-terms form, with the root-locus gain $K_{rlg} = 2.0$. The pole-zero map is illustrated in figure 18



Poles, Eqn (79):
0 , -3 +/- j2

Figure 18: Pole-zero map of system with a complex pole pair.

- The residue at pole p_2 is computed with the aid of figure 19

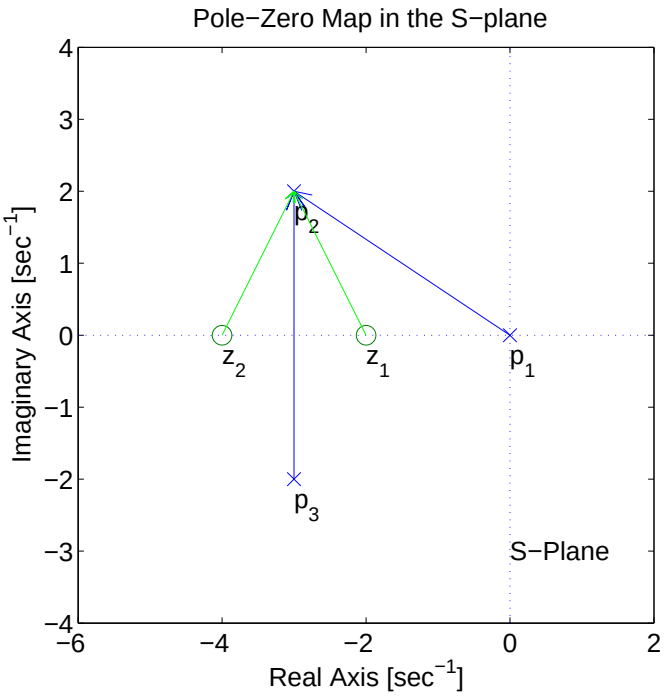


Figure 19: Graphical calculation of residue at pole p_2.

- Considering the vectors

$$C_2 = K_{rlg}\frac{\Pi \text{Vecs from zeros}}{\Pi \text{Vecs from poles}} = 2\,\frac{2.23\angle 116.56^o \quad 2.23\angle 63.43^o}{3.60\angle 146.31^o \quad 4\angle 90^o}$$

$$= 2\,\frac{5\angle 180^o}{14.42\angle 236.31^0} = 2\times 0.347\angle -56.31 = 0.694\angle -56.31$$

$$= 0.385 - j0.577$$

- Double checking with Matlab

```
>> N = [2 12 16];  D = [1 6 13];
>> [Residues, Poles] = residue(N, D)
      Residues =    0.3846 - 0.5769i
                    0.3846 + 0.5769i
                    1.2308

      Poles =      -3.0000 + 2.0000i
                   -3.0000 - 2.0000i
                    0
```

```
>> norm(Residues(1))
  ans =   0.6934
>> angle(Residues(1))*(180/pi)
  ans =  -56.3099
```

6 Interpreting pole location on the S-Plane

- We've seen how discovering the time-domain response of a system has two parts

  1. Finding the poles of the system
  2. Finding the residues at the poles

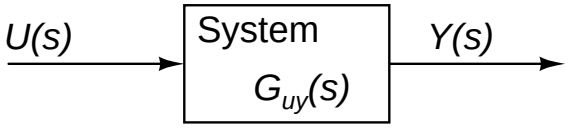
- And we've seen that each pole (or complex pole pair) gives a term in the partial fraction expansion

$$F(s) = \frac{C_1}{s - p_1} \stackrel{\mathcal{L}^{-1}}{\Longrightarrow} f(t) = C_1 e^{p_1 t} \tag{80}$$

- We've also seen where the terms C_i and p_i come from:
  - Exponents p_i are the poles, they come from finding the roots of $A(s)$
  - The C_i are the residues, found by one of our methods

- Considering Eqn (80), for each term the residue determines the magnitude, but the pole determines the exponential.
  - The magnitude is important,
  - But the exponential is much more important for characterizing the response of the system.

6.1 Modes of a system response



- Considering the response of system $G_{uy}(s)$. $Y(s)$ is given in sum-of-terms form as

$$Y(s) = G_{uy}(s)\,U(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \cdots \tag{81}$$

- And we've seen that each pole (or complex pole pair) gives a term in the partial fraction expansion

$$F(s) = \frac{C_1}{s-p_1} \overset{\mathcal{L}^{-1}}{\Longrightarrow} f(t) = C_1\,e^{p_1 t} \tag{82}$$

- We've also seen where the terms C_i and p_i come from:
  - Exponents p_i are the poles, they come from finding the roots of $A(s)$
  - The C_i are the residues, found by one of our methods
- Considering Eqn (82), for each term the residue determines the magnitude, but the pole determines the exponential.
  - The magnitude is important,
  - But the exponential is much more important for characterizing the response of the system.

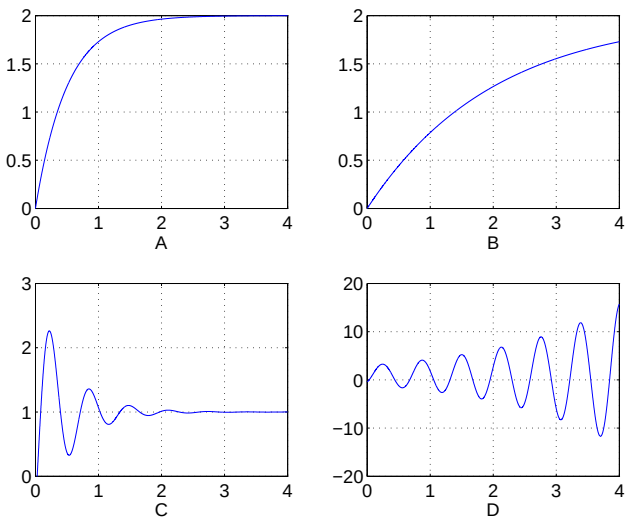
- Typically, one term in Eqn (81) comes from $U(s)$, the rest of the terms come from $G_{uy}(s)$

  – We call the terms coming from $G_{uy}(s)$ the ***modes*** of the system. The modes depend on the poles of the system.

  – The poles of $G_{uy}(s)$ give terms $y(t)$, no matter what input $U(s)$ is applied.

  – Several typical modes are seen in figure 20. The exponential part of the term conveys basic characteristics of the system: speed, damping and stability.

  – We can determine the characteristics of a mode by looking at where the corresponding pole of $G_{uy}(s)$ falls on the s-plane.



A: Fast, stable first-order mode

B: Slow stable mode

C: Stable, damped oscillation (second-order mode)

D: Unstable second-order mode

Figure 20: Characteristic responses of linear systems.

Note: For simplicity, the equations in this discussion are written considering first-order modes. But all points apply to both first- and second-order modes.

Definition a of Mode: Each real pole or complex pole pair in the transfer function of a system corresponds to an exponential term in the system response (such as Eqn (83)), and is called a mode.

- We say "a mode of the system," if one is talking about poles of $G_{uy}(s)$, or
- We say "a mode of the response," referring to poles of $Y(s)$ or terms in $y(t)$.
- Complex pole pairs introduce only one mode for each pole pair.

Example,

$$y(t) = \left(\frac{5}{6}\right) 1^{+}(t) + \left(\frac{5}{12}\right) e^{-6t} + \left(\frac{5}{4}\right) e^{+2t} \tag{83}$$

- Signal $y(t)$ has 3 modes, corresponding to the three real poles in $Y(s)$.
  - DC mode, corresponding to the pole at the origin and the $1^{+}(t)$ term.
    - With the unit step input, the residue corresponds to K_{DC} of the system.
  - The stable exponential mode, corresponding to the pole in the left-half-plane.
  - The unstable exponential mode, corresponding to the pole in the right-half-plane.

6.2 Regions of the S-plane

Since

$$\mathcal{L}^{-1}\left\{\frac{C_i}{s-p_i}\right\} = C_i e^{p_i t} \tag{84}$$

Much about the ith mode of the response is determined by where p_i lies on the S-place.

- First and foremost, the left-half-plane is stable, while the right-half-plane (RHP) is unstable
- If p_i is positive, $C_i e^{p_i t}$ grows towards infinity, which is an ***unstable*** response
- Considering Eqn (84), the S-plane can be divided into 3 regions:

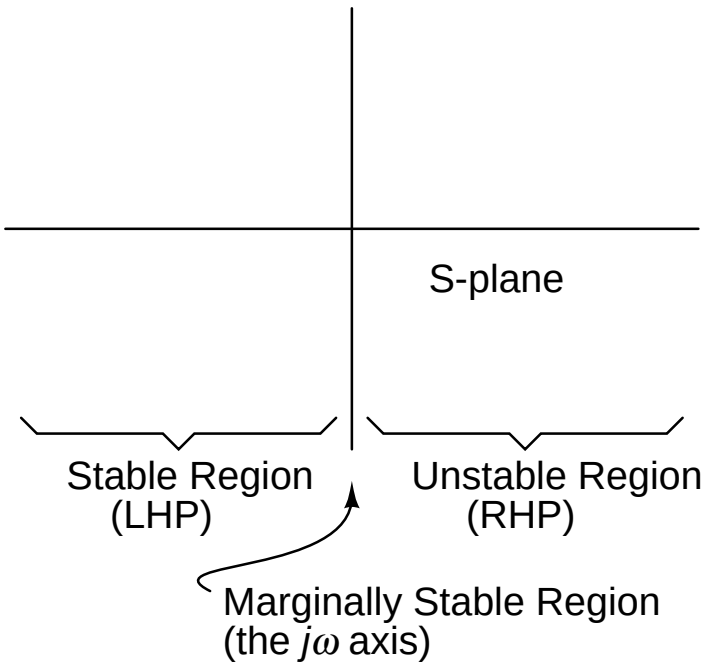


Figure 21: Regions of the S-plane.

7 Conclusions

In this part, we've seen

- Laplace transform properties and theorems
- Laplace transform pairs
- The inverse Laplace transform, and three methods to determine the residues of Partial Fraction Expansion
  - Plug and chug method
  - Cover-up method
  - Graphical method
- Modes of the response and pole locations on the S-plane
  - How modes correspond to poles and pole pairs, and
  - How the characteristics of the mode of the response corresponds the location of the pole and magnitude of the residue.
- The topics of this part will lead to tools to analyze and design control systems